Gianluca **Nordio**

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Work Experience

Robotize — *Denmark*

Robotics Software Engineer (full-time) — *Sep 2026 - present*

- Training and deploying state-of-the-art object detection and segmentation models for mobile robot navigation
- ROS, ROS 2, C++, Python, PyTorch, TensorRT, Jenkins, Simulation, Mobile robotics

Robotize — *Denmark*

Student AI Researcher — *Jul 2025 - Aug 2026*

- Trained and deployed object detection and segmentation models for robot navigation
- Made the company testing pipeline platform independent, so tests run across older and newer software versions
- Found more than 10 critical bugs
- Carried out my M.Sc. thesis on robust AMR localization in collaboration with the company

MARS Lab — *Denmark*

Student Researcher — *2025 - 2026*

- Developed RL policies for state-of-the-art autonomous racing drones competing in the A2RL autonomous drone championship
- Reimplemented and improved state-of-the-art RL methods for agile, time-optimal autonomous flight
- Closed the sim-to-real gap: deployed trained policies on edge hardware, handling real-world latency, noise and aerodynamics

Leo Club Italy — *Italy*

Software Engineer and Volunteer — *2022 - present*

- Built and maintain the website, reducing costs to 1% of the original
- Implemented an internal data portal for 3000 users
- Organized events and presentations with more than 150 participants each; raised more than 3000 euros for blind people

Education

Technical University of Denmark (DTU) — *Kgs. Lyngby, Denmark*

M.Sc. in Autonomous Systems — *Graduated Aug 2026*

- **Thesis** (with Robotize): Robust AMR localization in dynamic industrial environments. A 2D-lidar map updater that keeps AMCL localization reliable as warehouses change, validated on 100 simulated scenarios and a real robot [**code**]
- **Special project**: Reinforcement learning for autonomous drone navigation inside confined ballast tanks
- Qualified second among 21 teams in the RoboCup robotics competition; active member of RoboClub
- **Main courses**: Distributed Real-Time Systems, Perception for Autonomous Systems, Digital Control, Unmanned Autonomous Systems, Robotics, Building Dependable Robot Systems, Vision Based RL

University of Padua (UNIPD) — *Padua, Italy*

M.Sc. in Computer Engineering: AI and Robotics — *Graduated Oct 2026*

- Wrote the official book for the Deep Learning course and helped improve the Computer Vision course
- GPA: 3.96; Ambassador of the University of Padua
- **Main courses**: Deep Learning, Computer Vision, Artificial Intelligence, Big Data, Machine Learning, Operations Research, Quality Engineering

Top International Managers in Engineering — *Europe*

TIME Double Degree (DTU and UNIPD) — *Completed 2026*

- One of 10 students selected from the Engineering School of Padua (14,000 students); ranked second; awarded a 2-year grant

University of Padua (UNIPD) — *Padua, Italy*

B.Sc. in Computer Engineering — *2023*

- **Final score**: 110/110 with honors (GPA: 4.00); received a merit-based grant for STEM degrees
- **Thesis**: Experimental comparison between RGB-D and IMU sensors for Human Action Recognition [**paper**]
- **Main courses**: Embedded Systems, Computer Networks, Data Structures & Algorithms, Operating Systems, Computer Architecture, Control Theory, Electronics, Probability, Linear Algebra

Projects

- **Paris Hackathon winner** ({Tech: Europe}): HireMind, an AI interviewer. 1st in MakiPeople track, 2nd of 30+ teams - [video]
- **SLAM with a Marine Surface Vehicle and Turtlebot3** (C++, Python, ROS 2)
- **Robotic Arm following a Drawn Line from a Picture** (Python, OpenCV) - [video | article]
- **Parking Space Occupancy using Computer Vision** (C++, OpenCV, CMake) - [code | article]
- **OpenCV open-source contributor** (C++) - [merged PR]